

Functions of Several Variables

Quick Reference Cheat Sheet

1 Definitions & Limits

Function of n Variables: Let $S \subseteq \mathbb{R}^n$, then $f : S \rightarrow \mathbb{R}$ is a real-valued function of n variables.

Neighbourhood: A rectangular neighbourhood of $(a, b) \in \mathbb{R}^2$ is: $N(a, b) = \{(x, y) : a - h < x < a + h, b - k < y < b + k\}$ for $h, k > 0$.

Limit Point: (a, b) is a limit point (condensation point) of S if every neighbourhood contains an infinite number of points of S .

Limit ($\varepsilon - \delta$ Definition): $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = l$ if for every $\varepsilon > 0, \exists \delta > 0$ such that $|f(x, y) - l| < \varepsilon$ whenever $0 < |x - a| < \delta$ and $0 < |y - b| < \delta$.

Non-existence of Limit:

If we can find two paths $\phi_1(x)$ and $\phi_2(x)$ approaching (a, b) such that the limits of $f(x, \phi(x))$ are different, the simultaneous limit does not exist.

2 Continuity

A function f is continuous at $(a, b) \in S$ if:

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$$

3 Derivatives

Directional Derivative: At point $p(a, b)$ in the direction $u = (u_1, u_2)$:

$$D_u(p) = \lim_{h \rightarrow 0} \frac{f(p + hu) - f(p)}{h}$$

Note: Exists iff partial derivatives exist and are continuous at the point.

Partial Derivatives:

Directional derivatives in direction of axes:

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h} \quad (\text{along } x)$$

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h} \quad (\text{along } y)$$

4 Differentiability

$f(x, y)$ is differentiable at (a, b) if:

$$f(a + h, b + k) - f(a, b) = Ah + Bk + g(h, k)\sqrt{h^2 + k^2}$$

and $\lim_{(h,k) \rightarrow (0,0)} g(h, k) = 0$, where $A = f_x(a, b)$ and $B = f_y(a, b)$.

Key Relations:

- Differentiable $\implies$ Continuous.
- Continuous $\not\implies$ Differentiable.
- Continuous partial derivatives $\implies$ Differentiable.

5 Jacobian & Transformations

For transformation $f(x_1, \dots, x_n) = (\phi_1, \dots, \phi_m)$, the **Jacobian matrix** at $x = c$ is:

$$D(f(c)) = \left[\frac{\partial \phi_i}{\partial x_j} \right]_{m \times n}$$

Inverse Function Theorem: If $f'(x)$ is continuously differentiable and $[Df(c)]$ is invertible, then f is invertible in some neighbourhood of c .

6 Extreme Values

Necessary Condition: For $(a_1, \dots, a_n)$ to be an extreme value, partial derivatives must vanish. Stationary points solve:

$$f_{x_1} = 0, \quad f_{x_2} = 0, \quad \dots, \quad f_{x_n} = 0$$

Sufficient Condition (Second Derivative Test): Let (a, b) be a stationary point. Let:

$$A = f_{xx}(a, b), \quad B = f_{xy}(a, b), \quad C = f_{yy}(a, b)$$

Discriminant: $\Delta = AC - B^2$.

- **Local Minimum:** $\Delta > 0$ and $A > 0$.
- **Local Maximum:** $\Delta > 0$ and $A < 0$.
- **Saddle Point:** $\Delta < 0$ (neither max nor min).
- **Doubtful Case:** $\Delta = 0$ (needs further investigation).